

ASSIGNMENT 1

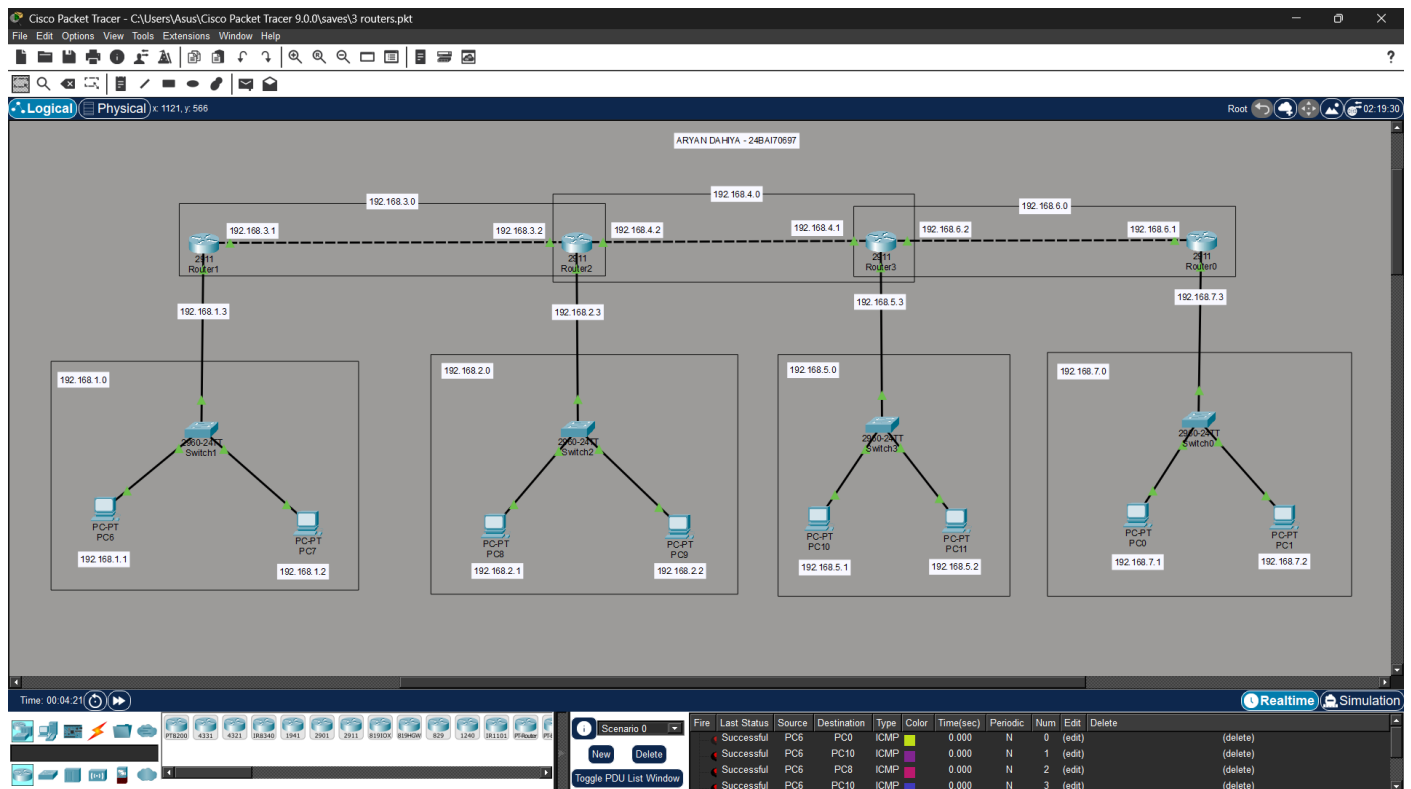
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QUESTION 1:

Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well.

SOLUTION:



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QUESTION 2:

What is OSI model? What are the different layers of OSI model? Write the functionality, Data format and Protocols used at each layer.

ANSWER 2:

OSI Model

The OSI (Open Systems Interconnection) Model is a conceptual framework developed by the International Organization for Standardization (ISO) to standardize communication between different computer systems over a network.

The main purpose of the OSI model is to divide the complete network communication process into different layers so that each layer performs a specific set of functions. Each layer provides services to the layer above it and receives services from the layer below it.

The OSI model consists of seven layers:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

When data is sent from a source device, it moves from the Application Layer to the Physical Layer. At the receiving device, it moves in the reverse direction, from the Physical Layer to the Application Layer.

1. Physical Layer – Layer 1

The Physical Layer is the lowest layer of the OSI model. It is responsible for the actual transmission of raw binary data in the form of electrical, optical, or radio signals through a physical communication medium.

Functions

- Transmits raw bits from one device to another.
- Defines the physical transmission medium such as copper cable, optical fibre, or wireless medium.
- Specifies connectors, voltage levels, signal types, and transmission rates.
- Determines whether transmission is simplex, half-duplex, or full-duplex.
- Defines the physical topology of the network.
- Converts binary data into signals suitable for transmission.

Data Format: Bits (0 and 1)

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Protocols/Standards

Ethernet Physical Layer Standards: Define how Ethernet signals are physically transmitted through cables or fibre.

IEEE 802.11 Physical Standards: Define radio transmission techniques used by Wi-Fi networks.

Bluetooth: A short-range wireless communication technology used for communication between nearby devices.

USB (Universal Serial Bus): Defines physical connections and signalling for communication between computers and peripheral devices.

RS-232: A serial communication standard used for communication between computers and serial devices.

2. Data Link Layer – Layer 2

The Data Link Layer provides reliable node-to-node communication between devices that are directly connected to each other. It receives raw bits from the Physical Layer and organizes them into meaningful units called frames.

It is also responsible for physical addressing using MAC (Media Access Control) addresses.

Functions

- Converts raw bits into frames.
- Provides node-to-node data transfer.
- Performs physical addressing using MAC addresses.
- Detects errors that occur during transmission.
- Performs flow control between directly connected devices.
- Controls access to the shared transmission medium.
- Adds a header and trailer to the data for proper transmission.

Data Format: Frame

Protocols

Ethernet (IEEE 802.3): A widely used LAN technology that defines framing, MAC addressing, and communication over Ethernet networks.

Wi-Fi (IEEE 802.11): Defines wireless LAN communication, including frame transmission and medium access.

PPP (Point-to-Point Protocol): Used to establish direct communication between two network nodes.

HDLC (High-Level Data Link Control): A bit-oriented protocol used for reliable communication over point-to-point and multipoint links.

3. Network Layer – Layer 3

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The Network Layer is responsible for transferring packets from the source network to the destination network. It provides logical addressing and determines the best path through which the packet should travel.

Routers mainly operate at the Network Layer.

Functions

- Provides logical addressing using IP addresses.
- Determines the best path between source and destination.
- Performs routing of packets between different networks.
- Forwards packets to their destination.
- Handles packet fragmentation when required.
- Enables communication between devices located on different networks.

Data Format: Packet

Protocols

IP (Internet Protocol): Provides logical addressing and is responsible for delivering packets from the source host to the destination host across networks. IPv4 and IPv6 are the two major versions.

ICMP (Internet Control Message Protocol): Used for network error reporting and diagnostic purposes. The ping utility uses ICMP messages.

OSPF (Open Shortest Path First): A routing protocol used by routers to determine efficient routes within an IP network.

RIP (Routing Information Protocol): A distance-vector routing protocol that uses hop count to select routes.

Note: ARP is commonly discussed alongside Layer 3 because it maps an IPv4 address to a MAC address, although its exact placement in the OSI model is often treated as between Layers 2 and 3.

4. Transport Layer – Layer 4

The Transport Layer is responsible for end-to-end communication between applications running on different devices. It divides large amounts of data into smaller units and ensures that the data is delivered appropriately to the destination application.

Depending on the protocol used, the Transport Layer can provide reliable or fast connectionless communication.

Functions

- Provides process-to-process communication.
- Performs segmentation and reassembly of data.
- Provides flow control.
- Provides error recovery when supported by the transport protocol.
- Uses port numbers to identify applications.

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- Supports connection-oriented and connectionless communication.
- Can provide reliable and ordered delivery of data.

Data Format:

TCP – Segment

UDP – Datagram

Protocols

TCP (Transmission Control Protocol): A connection-oriented and reliable transport protocol. It provides sequencing, acknowledgements, retransmission, flow control, and error recovery. It is commonly used by applications where reliable delivery is important.

UDP (User Datagram Protocol): A connectionless transport protocol that sends data without establishing a connection. It has lower overhead than TCP but does not guarantee delivery, ordering, or retransmission.

5. Session Layer – Layer 5

The Session Layer is responsible for establishing, maintaining, and terminating communication sessions between two applications. A session represents an active communication between applications on different systems.

Functions

- Establishes communication sessions.
- Maintains active sessions.
- Terminates sessions after communication is completed.
- Controls the dialogue between communicating applications.
- Provides synchronization using checkpoints.
- Helps resume communication from a checkpoint after an interruption.

Data Format: Data

Protocols/Technologies

NetBIOS: Provides services that allow applications on different computers to communicate over a network and is commonly associated with session management.

RPC (Remote Procedure Call): Allows a program on one computer to execute a procedure or function on another computer over a network.

6. Presentation Layer – Layer 6

The Presentation Layer is responsible for the representation and formatting of data. Different computer systems may represent information differently; therefore, this layer ensures that data sent by one system can be correctly interpreted by another.

It is sometimes called the translation layer of the OSI model.

Functions

- Converts data from one format to another.

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- Performs character encoding and decoding.
- Performs encryption and decryption.
- Performs data compression and decompression.
- Ensures that transmitted information is represented in a suitable format.

Data Format: Data

Protocols/Formats

TLS (Transport Layer Security): Provides encryption and authentication for secure communication. In practical TCP/IP implementations, TLS does not map perfectly to a single OSI layer but is commonly associated with presentation-layer functions.

SSL (Secure Sockets Layer): The older predecessor of TLS that provided encrypted communication. SSL is now obsolete and has been replaced by TLS.

JPEG: A standard format used for representing and compressing digital images.

MPEG: A family of standards used for encoding and compressing audio and video.

ASCII: A character encoding standard used to represent text characters in computers.

7. Application Layer – Layer 7

The Application Layer is the highest layer of the OSI model. It provides network services that are used by end-user applications. It acts as the interface through which software applications access network communication services.

For example, when a user opens a website, sends an email, or transfers a file, application-layer protocols are involved.

Functions

- Provides network services to application programs.
- Supports web browsing.
- Supports electronic mail services.
- Provides file transfer services.
- Provides name resolution.
- Supports remote access and network resource sharing.
- Acts as an interface between user applications and network services.

Data Format: Data

Protocols

HTTP (Hypertext Transfer Protocol): Used to transfer web pages and other web resources between web clients and web servers.

HTTPS (Hypertext Transfer Protocol Secure): HTTP communication protected using TLS, providing confidentiality, integrity, and server authentication.

FTP (File Transfer Protocol): Used for transferring files between computers over a network.

SMTP (Simple Mail Transfer Protocol): Used mainly for sending and relaying email messages.

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POP3 (Post Office Protocol Version 3): Used by email clients to retrieve emails from a mail server, traditionally by downloading messages.

IMAP (Internet Message Access Protocol): Used to access and manage email while keeping messages synchronized with the mail server.

DNS (Domain Name System): Converts human-readable domain names, such as example.com, into IP addresses that computers use for communication.

QUESTION 3:

What are the different Error Detection methods, Explain each with the help of proper example?

Error Detection is the process of identifying whether the received data contains any errors. It enables the receiver to determine if the transmitted data is correct. If an error is detected, the receiver requests the sender to retransmit the data. Error detection improves the reliability and accuracy of data communication.

The most commonly used error detection methods are:

1. Parity Check
2. Checksum
3. Cyclic Redundancy Check (CRC)
4. Longitudinal Redundancy Check (LRC)

1. Parity Check

Definition

Parity Check is the simplest and most commonly used error detection technique. In this method, an additional bit called the **parity bit** is added to the original data before transmission. The parity bit is selected so that the total number of **1s** in the transmitted data becomes either **even** or **odd**.

There are two types of parity:

- **Even Parity:** The total number of 1s in the transmitted data must be even.
- **Odd Parity:** The total number of 1s in the transmitted data must be odd.

At the receiver's end, the parity is checked. If the parity does not match the expected value, the receiver concludes that an error has occurred.

Working

1. Count the number of 1s in the data.
2. Add the parity bit.
3. Transmit the data.
4. The receiver counts the number of 1s again.

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5. If parity is incorrect, an error is detected.

Example

Original Data:

1011001

Number of 1s = **4 (Even)**

Even Parity

Since the number of 1s is already even, the parity bit is **0**.

Transmitted Data

10110010

Suppose the receiver receives

10110110

Now the total number of 1s becomes **5**, which is odd.

Since even parity was expected, the receiver detects an error.

Advantages

- Very simple to implement.
- Requires only one additional bit.
- Fast error detection.

Disadvantages

- Detects only single-bit errors.
- Cannot detect all multiple-bit errors.
- Cannot identify the location of the error.

2. Checksum

Definition

Checksum is an error detection technique in which the sender divides the data into fixed-size blocks (such as 8-bit or 16-bit words). All blocks are added together using binary addition. The **one's complement** of the final sum is calculated and transmitted as the checksum.

The receiver performs the same calculation. If the newly calculated checksum matches the received checksum, the data is accepted. Otherwise, an error is detected.

Checksums are widely used in communication protocols because they are simple and efficient.

Working

1. Divide the data into equal-sized blocks.
2. Add all blocks using binary addition.
3. Calculate the one's complement of the result.
4. Attach the checksum to the data.

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5. The receiver repeats the same calculation.
6. If the result is correct, the data is accepted; otherwise, an error is detected.

Example

Suppose two 8-bit data blocks are

10101010

11001100

Binary Addition

10101010

11001100

101110110

The carry is added back (end-around carry), and the one's complement of the final sum is calculated.

Suppose the checksum obtained is

00100100

The sender transmits

Data + Checksum

The receiver performs the same calculation.

- If the final result is all **1s**, the data is accepted.
- Otherwise, the receiver detects an error.

Applications

- Internet Protocol (IP)
- TCP
- UDP

Advantages

- Easy to implement.
- More reliable than parity checking.
- Detects many transmission errors.
- Requires less computation than CRC.

Disadvantages

- Cannot detect all error patterns.
- Less reliable than CRC.

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3. Cyclic Redundancy Check (CRC)

Definition

Cyclic Redundancy Check (CRC) is one of the most powerful and reliable error detection techniques used in modern communication systems. In CRC, the sender treats the data as a binary polynomial and divides it by a predefined **Generator Polynomial (Divisor)** using **Modulo-2 division** (binary division without carry or borrow).

The remainder obtained after the division is called the **CRC code** or **Frame Check Sequence (FCS)**. This remainder is appended to the original data before transmission.

At the receiver's end, the same division is performed using the same generator polynomial.

- If the remainder is **0**, the received data is considered correct.
- If the remainder is **non-zero**, an error is detected.

CRC is capable of detecting single-bit errors, multiple-bit errors, and burst errors with very high accuracy.

Working

1. Select a generator polynomial.
2. Append zeros to the data equal to the degree of the polynomial.
3. Perform modulo-2 division.
4. Obtain the remainder (CRC).
5. Append the CRC to the data.
6. The receiver performs the same division.
7. If the remainder is zero, the data is accepted.

Example

Original Data

110101

Generator Polynomial

1011

After modulo-2 division, assume the remainder obtained is

110

The transmitted frame becomes

110101110

At the receiver,

$110101110 \div 1011$

If the remainder obtained is

000

No error has occurred.

If the remainder is

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101

The receiver detects an error.

Applications

- Ethernet Networks
- Wi-Fi Networks
- USB Communication
- Hard Disk Drives
- Optical Discs

Advantages

- Detects single-bit errors.
- Detects multiple-bit errors.
- Excellent at detecting burst errors.
- Highly reliable.
- Most widely used error detection technique.

Disadvantages

- More complex than parity and checksum.
- Requires polynomial division hardware or software.

4. Longitudinal Redundancy Check (LRC)

Definition

Longitudinal Redundancy Check (LRC) is an error detection technique in which data is arranged in the form of rows and columns. A parity bit is generated for every column, producing an additional row called the **LRC character**.

Both the original data and the LRC character are transmitted to the receiver. The receiver recalculates the column parity and compares it with the received LRC.

If any column parity is incorrect, an error is detected.

LRC provides better error detection than simple parity because it checks multiple bits simultaneously.

Working

1. Arrange data in rows.
2. Generate parity for every column.
3. Form the LRC row.
4. Transmit data along with the LRC.
5. The receiver recalculates the parity.
6. If parity differs, an error is detected.

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Example

Suppose four data bytes are

1010

1100

1001

1111

Arrange them as follows:

Byte Data

1 1010

2 1100

3 1001

4 1111

Column-wise parity gives

1000

This parity row is transmitted as the **LRC**.

If the receiver calculates a different parity row, it concludes that an error has occurred.

Applications

- Magnetic Tape Storage
- Serial Communication
- Point-of-Sale (POS) Systems
- Industrial Communication Systems

Advantages

- Better than simple parity.
- Detects many multiple-bit errors.
- Simple implementation.

Disadvantages

- Less reliable than CRC.
- Cannot detect every possible error pattern.